ESLSCA UNIVERSITY

MBA PROGRAM

ACCOUNTING FOR MANAGERS

MANAGERIAL ACCOUNTING [PART TWO]

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LECTURE THTREE :DECISION MAKING ANALYSIS

TASK ONE: Keep or Drop a Product Line

The Cook Corporation has two divisions--East and West. The divisions have the following revenues and expenses:

	East	West
Sales	$ 500,000	550,000
Variable costs	200,000	275,000
Traceable fixed costs	150,000	180,000
Allocated common corporate costs	305000	170,000
Net operating income (loss)	$ 15000	(75000)

The management of Cook is considering the elimination of the West Division. If the West Division were eliminated, its traceable fixed costs could be avoided. Total common corporate costs would be unaffected by this decision.

a- Given these data, the elimination of the West Division would result in an overall company net operating income (loss) of:

A) $15,000 B) ($155,000) C) ($75,000) D) ($60,000)

b- Calculate the breakeven sales for the West Division

CM%= CM/Sales =275000/550000=50%

Break even for West Division is to cover its Traceable Fixed =180000/50%=360000

TASK TWO: Keep or Drop a Product Line

Vanik Corporation currently has two divisions which had the following operating results for last year:

		Cork Division	Rubber Division
Sales	$	600,000	350,000
Variable costs		250,000	220,000
Contribution margin		350,000	130,000
Traceable fixed costs		160,000	110,000
Segment margin		190,000	20,000
Allocated common corporate fixed costs		80,000	45,000
Net operating income (loss)	$	110,000	(25,000)

Because the Rubber Division sustained a loss, the president of Vanik is considering the elimination of this division. All of the division's traceable fixed costs could be avoided if the division was dropped. None of the allocated common corporate fixed costs could be avoided.

a-If the Rubber Division was dropped at the beginning of last year, the financial advantage (disadvantage) to the company for the year would have been:

A) ($20,000) B) $20,000 C) $25,000 D) ($25,000)

b- Calculate the breakeven sales for the rubber division

CM%= 130000/350000=

Breakeven dollars =110,000/CM%

TASK THREE: Keep or Drop a Product Line

Mr. Earl Pearl, accountant for Margie Knall Co., Inc., has prepared the following product-line income data:

		Product		
	Total	A	B	C
Sales	$100,000	$50,000	$20,000	$ 30,000
Variable expenses	60,000	30,000	10,000	20,000
Contribution margin	40,000	20,000	10,000	10,000
Fixed expenses:				
Rent	5,000	2,500	1,000	1,500
Depreciation	6,000	3,000	1,200	1,800
Utilities	4,000	2,000	500	1,500
Supervisors' salaries	5,000	1,500	500	3,000
Maintenance	3,000	1,500	600	900
Administrative expenses	10,000	3,000	2,000	5,000
Total fixed expenses	33,000	13,500	5,800	13,700
Net operating income	$ 7,000	$ 6,500	$ 4,200	$(3,700)

The following additional information is available:

- The factory rent of $1,500 assigned to Product C is avoidable if the product were dropped.
- The company's total depreciation would not be affected by dropping C.
- Eliminating Product C will reduce the monthly utility bill from $1,500 to $800.
- All supervisors' salaries are avoidable.
- If Product C is discontinued, the maintenance department will be able to reduce monthly expenses from $3,000 to $2,000.
- Elimination of Product C will make it possible to cut two persons from the administrative staff; their combined salaries total $3,000.

  Required:

  Prepare an analysis showing whether Product C should be eliminated.

Traceable Fixed Costs = 1500+ 700+3000+900+3000= 9100

Segment Margin = 10,000 – 9100= 900 + KEEP

TASK FOUR Make or Buy Decisions

Kirsten Corporation makes 100,000 units per year of a part called a B345 gasket for use in one of its products. Data concerning the unit production costs of the B345 gasket follow:

Direct materials	$0.15
Direct labor	0.10
Variable manufacturing overhead	0.13
Fixed manufacturing overhead	0.24
Total manufacturing cost per unit	$0.62

An outside supplier has offered to sell Kirsten Corporation all of the B345 gaskets it requires. If Kirsten Corporation decided to discontinue making the B345 gaskets, 25% of the above fixed manufacturing overhead costs could be avoided. Assume that direct labor is a variable cost.

Required:

a. Assume Kirsten Corporation has no alternative use for the facilities presently devoted to production of the B345 gaskets. If the outside supplier offers to sell the gaskets for $0.46 each, should Kirsten Corporation accept the offer? Fully support your answer with appropriate calculations.

b. Assume that Kirsten Corporation could use the facilities presently devoted to production of the B345 gaskets to expand production of another product that would yield an additional contribution margin of $10,000 annually. What is the maximum price Kirsten Corporation should be willing to pay the outside supplier for B345 gaskets?

Answer:

a. The analysis of the alternatives follows below:

	Make	Buy
Purchase cost		$0.46
Direct materials	$0.15	
Direct labor	0.10	
Variable manufacturing overhead	0.13	
Fixed manufacturing overhead*	0.06	
Total cost	$0.44	$0.46

* 25% × $0.24

The company should make the part rather than buy it from the outside supplier because it costs $0.02 less under that alternative.

b. The maximum acceptable price is $0.54 because that is the cost to the company of making the part itself when the opportunity cost is included:

Total cost of making the part internally	$0.44
Opportunity cost per unit ($10,000 ÷ 100,000)	0.10
Total	$0.54

TASK FIVE: Keep or Drop a Product Line

Epsilon Co. can produce a unit of product for the following costs:

Direct material	$ 8
Direct labor	24
Overhead	40
Total product costs per unit	$ 72

An outside supplier offers to provide Epsilon with all the units it needs at $60 per unit. If Epsilon buys from the supplier, the company will still incur 40% of its overhead. Epsilon should choose to:

A) Buy since the relevant cost to make it is $72.

B) Make since the relevant cost to make it is $56.

C) Buy since the relevant cost to make it is $48.

D) Make since the relevant cost to make it is $48.

E) Buy since the relevant cost to make it is $56.

Answer: Make =8+24 +24 = 56 Vs Buy 60

TASK SEVEN: Keep or Drop a Product Line

Frederick Co. is thinking about having one of its products manufactured by an outside supplier.
Currently, the cost of manufacturing 5,000 units is:

Direct material	$ 62,000
Direct labor	47,000
Variable factory overhead	38,000
Factory overhead	52,000

If Frederick can buy 5,000 units from an outside supplier for $130,000, it should:

A) Make the product because current factory overhead is less than $130,000.
B) Make the product because the cost of direct material plus direct labor of manufacturing is less than $130,000.
C) Make the product because factory overhead is a sunk cost.
D) Buy the product because total fixed and variable manufacturing costs are greater than $130,000.
E) Buy the product because the total incremental costs of manufacturing are greater than $130,000.

Answer: Make 62000 +47000 +38000=147,000 Vs Buy 130,000 (E)

LECTURE FOUR: COSTING

TASK ONE

Selected information from the budget of the Singh Corp. at the beginning of the year follows:

Estimated factory overhead	$132,000
Estimated direct labor hours	55,000 hours
Estimated machine hours………	41,250 hours
Estimated direct labor cost	$825,000
Actual factory overhead incurred during the year	$144,000

Calculate the predetermined overhead rate if the company uses the following as a basis:

(a) Direct labor hours.

(b) Direct labor cost.

(c) Machine hours.

Answers

Answer:

(a) $132,000/55,000 = $2.40 per direct labor hour

(b) $132,000/$825,000 = 16% of direct labor cost

(c) $132,000/41,250 = $3.20 per machine hour

TASK TWO

Green Cabinets is a custom cabinet builder. They recently completed a set of kitchen cabinets (Job Number 1478), as summarized below:

Job Number:1478

Date Started: 4/07/20x8

Direct material cost 890 Direct labor hours = 51 hours Direct labor hour =$10/hour Green Cabinets applies overhead to jobs at a rate of $12 per direct labor hour Required: Complete the following table:

Direct Material Cost	$	890
Direct Labor Cost		510
Applied Manufacturing Overhead		51x12=612
Total Cost		

TASK THREE:

Aztec Industries produces bread which goes through two operations, mixing and baking, before it is ready to be packaged. Next year's expected costs and activities are shown below.

	Mixing	Baking
Direct labor hours	400,000 DLH	80,000 DLH
Machine hours	800,000 MH	800,000 MH
Overhead costs	$600,000	$400,000

Compute

a. Aztec's departmental overhead rate for the mixing department based on direct labor hours.
b. Aztec's departmental overhead rate for the mixing department based on machine hours.
c. Aztec's departmental overhead rate for the baking department based on direct labor hours.

d. Aztec's departmental overhead rate for the baking department based on machine hours.

Answer

a. Compute Aztec's departmental overhead rate for the mixing department based on direct labor hours.
Explanation: $600,000/400,000 DLH = $1.50 per DLH

b. Compute Aztec's departmental overhead rate for the mixing department based on machine hours.
Explanation: $600,000/800,000 MH = $0.75 per MH

c. Compute Aztec's departmental overhead rate for the baking department based on direct labor hours.
Explanation: $400,000/80,000 DLH = $5.00 per DLH

d. Compute Aztec's departmental overhead rate for the baking department based on machine hours.
Explanation: $400,000/800,000 MH = $0.50 per MH

TASK FOUR

Carlson Company has two departments. Factory overhead costs are applied based on direct labor cost in Department A and machine hours in Department B.
The following information is available:

Budgeted Costs	Dept. A	Dept. B
Direct labor cost	$150,000	$165,000
Machine hours	50,000	$20,000
Factory overhead cost	$225,000	$180,000

Actual data for Job #10 are as follows:

Actual Costs	Dept. A	Dept. B
Direct materials requisitioned	$10,000	$16,000
Direct labor cost	$11,000	$14,000
Machine hours	5,000	3,000

Required:

Compute the budgeted factory overhead rate for Department A.

POHR (A)= 225000/150,000 = $1.5/DLCost

Compute the budgeted factory overhead rate for Department B.

POHR (B) = 180000 /20000= $9/Mhr

What is the total overhead cost for Job #10?

Applied OH = (1.5 x 11000) + (9 x 3000)= 43500

If Job #10 consists of 50 units of product, what is the unit cost of this job?

Total Cost =A = 10000 +11000 + 16500

+B =16000 +14000 + 27000 = 94500

Cost /unit = 94500/ 50 =1890

Assuming that the co sets a mark up at 10%. What is the selling price

Selling price = 1890 x 110%= 2079

TASK FIVE:

Taylor Corp. uses a job order costing system and worked only on Job 101 during the current period. Job 101 was sold for $460,000. The following information pertains to costs incurred for Job 101.

Direct materials	$90,000
Indirect materials	$30,000
Direct labor	$130,000
Indirect labor	$75,000
Depreciation of machinery	$10,000
Factory supplies	$8,000
Overhead rate	90% of direct labor

After adjusting for the amount of over- or underapplied overhead, determine the amount of gross profit earned during the year.

Answer:

Actual Overhead = 30,000 +75000+ 10000 + 8000= 123000

Applied Overhead = 90% x 130,000 = 117000

So the company underapplied OH by 123000 -117000 = $6000

Cost = DM 90,000 + DL 130,000 + OH 123000= 343000

Gross Profit = Sales 460,000 – COGS 343000 = $117,000

TASK SIX

The predetermined overhead rate for Foster, Inc., is based on estimated direct labor costs of $400,000 and estimated factory overhead of $500,000.

Actual costs incurred were:

Direct materials	$240,000
Direct labor	410,000
Indirect materials	55,000
Indirect labor	125,000
Sales commissions	55,000
Factory depreciation	170,000
Property taxes, factory	15,000
Factory utilities	35,000
Advertising	62,500
Factory equipment rental	110,000

a)Calculate the predetermined overhead rate and calculate the overhead applied during the year.

b)Determine the amount of over- or underapplied overhead

Answer:

Predetermined overhead rate = $500,000/$400,000 = 125% of direct labor cost
Overhead applied = $410,000 * 125% = $512,500

Actual overhead:	
Indirect materials	$ 55,000
Indirect labor	125,000
Factory depreciation	170,000
Property taxes, factory	15,000
Factory utilities	35,000
Factory equipment rental	110,000
Total actual overhead	$510,000
Overhead applied	512,500
Overapplied overhead	$ 2,500

TASK SEVEN

The Lynn Company uses a normal job-costing system at its Minneapolis plant. The plant has a machining department and an assembly department. Its job-costing system has two direct-cost categories (direct materials and direct manufacturing labor) and two manufacturing overhead cost pools (the machining department overhead, allocated to jobs based on machine-hours, and the assembly department overhead, allocated to jobs based on actual direct manufacturing labor **costs).** The 2021 budget for the plant is as follows: (beginning 2021)

	Machining Department	Assembly Department
Manufacturing Overhead	1,800,000	3,600,000
Direct Manufacturing labor costs	1,400,000	**2,000,000**
Direct manufacturing labor hours	1,000	200,000
Machine Hours	50,000	200,000

1-Compute the budgeted manufacturing overhead rate for each department.

During February, the job-cost record for Job 494 contained the following:

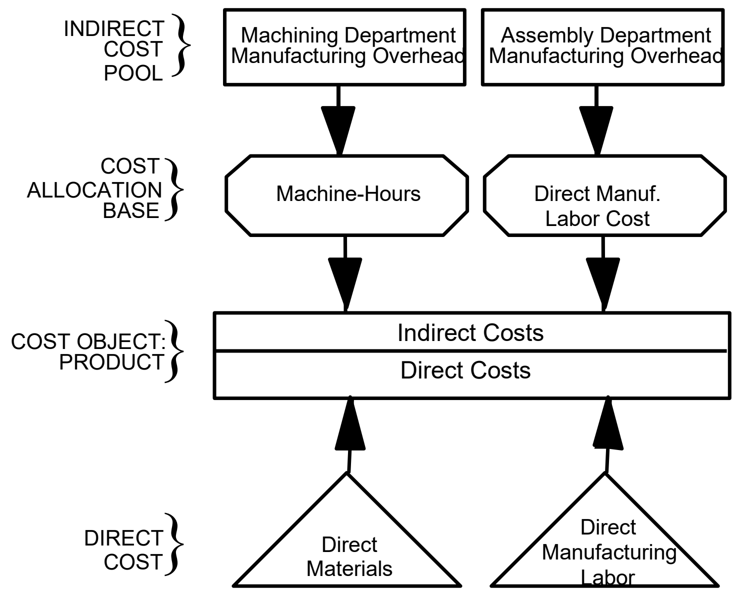
	Machining Department	Assembly Department
Direct Materials Used	45,000	70,000
Direct Manufacturing labor costs	14,000	15,000
Direct manufacturing labor hours	1400	1500

2-Compute the total manufacturing costs for Job 494.

3. At the end of 2021, the actual manufacturing overhead costs were $2,100,000 in machining and $3,700,000in assembly. Assume that 55,000 actual machine-hours were used in machining and that actual direct manufacturing labor costs in assembly were $2,200,000. Compute the over- or under-allocated manufacturing overhead for each department.

Answer:

An overview of the product costing system is



Budgeted manufacturing overhead divided by allocation base:

Machining overhead $\frac{\$1{,}800{,}000}{50{,}000}$ = \$36 per machine-hour

Assembly overhead: $\frac{\$3{,}600{,}000}{\$2{,}000{,}000}$ = 180% of direct manuf. labor costs

2.

Machining department, 2,000 hours × \$36	\$72,000
Assembly department, 180% × \$15,000	27,000
Total manufacturing overhead allocated to Job 494	\$99,000

3.

	Machining	**Assembly**
Actual manufacturing overhead	\$2,100,000	\$ 3,700,000
Manufacturing overhead allocated,		
55,000 × \$36	1,980,000	—
180% × \$2,200,000	—	3,960,000
Under allocated (Overallocated)	\$ 120,000	\$ (260,000)

TASK EIGHT

A company identified the following partial list of activities, costs, and activity drivers expected for the next year:

Activity	Expected Costs	Cost Driver
Extrusion costs	$83,600	Number batches made
Handling costs	$8,800	Number of orders filled
Packaging costs	$40,500	Number of units made

	Product A	Product B
Production volume	750,000 units	600,000 units
Batches made	200 batches	750 batches
Orders filled	75	200

a. Calculate activity rates for each of the three activities using activity-based costing (ABC).
b. How much overhead in total will be assigned to the Product A line using activity-based costing?

A company identified the following partial list of activities, costs, and activity drivers expected for the next year:

Estimated = estimated od / estimated cost driver

Activity	Expected Costs	Cost Driver	Activity rate
Extrusion costs	$83,600 $	Number batches made	83600/950= 88/batch
Handling costs	$8,800	Number of orders filled	8800/275= 32/order
Packaging costs	$40,500	Number of units made	40500/1350000= 0.03/unit

actual

	Product A	Product B	Total	Activity rate
Production volume	750,000 unitsx0.03	600,000 units x0.03	1350,000 units	
Batches made	200 batches x88	750 batches x88	950 batches	
Orders filled	75 ordersx32	200 orders x32	275 orders	

Calculate activity rates for each of the three activities using activity-based costing (ABC).

a. How much overhead in total will be assigned to the Product A line using activity-based costing?

Answer a and b

Extrusion activity: ($88/batch)(200) = $17,600
Handling costs: ($32/order)(75 orders) = $2,400
Packaging activity: ($.03/unit)(750,000 units) = $22,500
Total overhead cost = $42,500

TASK NINE

Fischer Company identified the following activities, costs, and activity drivers:

Activity	Expected Costs	Expected Activity
Handling parts	$425,000	25,000 parts in stock
Inspecting product	$390,000	940 batches
Processing purchase orders	$220,000	440 orders
Designing packaging	$230,000	5 models

a. Compute a plantwide overhead rate assuming the company assigns overhead based on 70,000 budgeted direct labor hours (Round to two decimals).

b. Compute separate rates for each of the four activities using the activity-based costing.

Answer:
a. ($425,000 + $390,000 + $220,000 + $230,000)/70,000 DLH = $18.07 per DLH
b. Handling: $425,000/25,000 parts = $17/part
Inspecting: $390,000/940 batches = $414.89/batch
Processing: $220,000/440 orders = $500/order
Designing: $230,000/5 models = $46,000/model

TASK TEN

Automotive Products (AP) designs and produces automotive parts. In 2021, actual variable manufacturing overhead is **$308,600**. AP's simple costing system allocates variable manufacturing overhead to its three customers based on machine-hours and prices its contracts based on full costs. One of its customers has regularly complained of being charged noncompetitive prices, so AP's controller Devon Smith realizes that it is time to examine the consumption of overhead resources more closely. He knows that there are three main departments that consume overhead resources: design, production, and engineering. Interviews with the department personnel and examination of time records yield the following detailed information.

	Cost driver	Variable Manufacturing OH	Usage of cost drivers by customer contract			
			United motors	Holden motors	Leland vehicle	
Design	CAD design hours	$39,000	110	200	80	
Engineering	Engineering hours	29,600	70	60	240	
Production	Machine hours	240,000	120	2,800	1,080	
Total		$ 308,600				

Required:

1. Compute the manufacturing overhead allocated to each customer in 2021 using the simple costing system that uses machine-hours as the allocation base.

2. Compute the manufacturing overhead allocated to each customer in 2021 using department-based manufacturing overhead rates.

3. Comment on your answers in requirements 1 and 2. Which customer do you think was complaining about being overcharged in the simple system?

1. Compute the manufacturing overhead allocated to each customer in 2021 using the simple costing system that uses machine-hours as the allocation base.
   308600/4000 = 77.15
   UM= 120 x 77.15=9258 under costed
   HM= 2800 x 77.15=216020 over costed
   LV = 1080 x77.17= 83322 under costed
2. Compute the manufacturing overhead allocated to each customer in 2021 using department-based manufacturing overhead rates.
   Design =39000/390= $100/design hour
   Engineering = 29600 / 370 = 80/engineering hour
   Machining = 240000/4000 = 60 /Mhr
3. Comment on your answers in requirements 1 and 2. Which customer do you think was complaining about being overcharged in the simple system? HM
   If the new department-based rates are used to price contracts, which customer(s) will be unhappy? UM and LV How would you respond to these concerns?

TASK ELEVEN

The job costing system at Sheri's Custom Framing has five indirect cost pools (purchasing, material handling, machine maintenance, product inspection, and packaging). The company is in the process of bidding on two jobs: Job 215, an order of 15 intricate personalized frames, and Job 325, an order of 6 standard personalized frames. The controller wants you to compare overhead allocated under the current simple job-costing system and a newly designed activity-based job-costing system. Total budgeted costs in each indirect cost pool and the budgeted quantity of activity driver are as follows.

	Budgeted Overhead	Activity driver	Budgeted quantity of activity driver	Predetermined Activity Rate
Purchasing	$35,000	Purchase orders produced	2,000	
Materials handling	43,750	Material moves	5,000	

Machine Maintenance	118,650	Machine hours	10,500	
Product inspection	9,450	Inspections	1,200	
Packaging	19950	Units produced	3,800	
	226,800			

Information related to Job 215 and Job 325 follows. Job 215 incurs more batch-level costs because it uses more types of materials that need to be purchased, moved, and inspected relative to Job 325.

	Job 215	Job 325
Number of purchase orders	25	8
Number of material moves	10	4
Machine-hours	40	60
Number of inspections	9	3
Units produced	15	6

Required:

1. Compute the total overhead allocated to each job under a simple costing system, where overhead is allocated based on machine-hours.
2. Compute the total overhead allocated to each job under an activity-based costing system using the appropriate activity drivers.
3. Explain why Sheri's Custom Framing might favor the ABC job-costing system over the simple job-costing system, especially in its bidding process.

	Budgeted Overhead	*Activity driver*	*Budgeted quantity of activity driver*	
Purchasing	*$35,000*	*Purchase orders produced*	*2,000*	*35000/2000= 17.5*
Materials handling	*43,750*	*Material moves*	*5,000*	*43750/5000= 8.75*
Machine Maintenance	*118,650*	*Machine hours*	*10,500*	*118650/10500= 11.30*
Product inspection	*9,450*	*Inspections*	*1,200*	*9450/1200= 7.87*

Packaging	*19950*	*Units produced*	*3,800*	*19950 /3800= 5.25*
	226,800			

a. Compute the total overhead allocated to each job under a simple costing system, where overhead is allocated based on machine-hours.
   POHR = 226800 /10500 = $21.6
   OH JOB 215 = 21.6 x 40= 864
   OH JOB 325 = 21.6 x 60= 1296
b. Compute the total overhead allocated to each job under an activity-based costing system using the appropriate activity drivers.

	Job 215	Job 325
Number of purchase orders	*25 x17.5*	*8 x17.5*
Number of material moves	*10x8.75*	*4 x 8.75*
Machine-hours	*40 x11.3*	*60 x 11.3*
Number of inspections	*9 x7.875*	*3 x 7.875*
Units produced	*15x 5.25*	*6x 5.25*
ABC	*1126*	*908*

3. Explain why Sheri's Custom Framing might favor the ABC job-costing system over the simple job-costing system, especially in its bidding process.

Job 215 was under-costed, the ABC revealed higher overhead allocated to this job. On the other hand, job 325 was over costed and this could reduce the sales for this product

LECTURE FIVE: BUDGETING

1-Aloan Co. provides the following sales forecast for the next three months:

	January	February	March
Sales units	3,000	4,200	5,000

The company wants to end each month with ending finished goods inventory equal to 10% of the next month's sales. Finished goods inventory on December 31 is 300 units. The budgeted production units for February are

A) 5,000 units.
B) 4,200 units.
C) 4,700 units.
D) 4,120 units.
E) 4,280 units.

2) Flack Corporation, a merchandiser, provides the following information for its December budgeting process:

The November 30 (Dec 1) inventory was 1,800 units.
Budgeted sales for December are 4,000 units.
Desired December 31 inventory is 2,840 units.
Budgeted purchases for December are:

A) 5,040 units.
B) 1,240 units.
C) 6,840 units.
D) 4,000 units.
E) 5,800 units.

3- A department store has budgeted sales of 12,000 men's suits in September. Management wants to have 6,000 suits in inventory at the end of the month to prepare for the winter season. Beginning inventory for September is expected to be 4,000 suits. What is the dollar amount of the purchase of suits if each suit has a cost of $75.

A) $750,000.
B) $900,000.
C) $1,050,000.
D) $1,200,000.
E) $1,350,000.

4) A sporting equipment store expects to purchase $8,000 of ski boots in October. The store had $2,000 of ski boots in merchandise inventory at the beginning of October, and expects to have $3,000 of ski boots in merchandise inventory at the end of October to cover part of anticipated November sales. What is the budgeted cost of goods sold for October?

A) $5,000.
B) $7,000.
C) $8,000.
D) $9,000.
E) $10,000.

5) Masterson Company's budgeted production calls for 56,000 liters in April and 52,000 liters in May of a key raw material that costs $1.85 per liter. Each month's ending raw materials inventory should equal 30% of the following month's budgeted materials. The April 1 inventory for this material is 16,800 liters. What is the budgeted materials need in liters for April?

A) 71,600 liters.
B) 39,200 liters.
C) 57,600 liters.
D) 56,000 liters.
E) 54,800 liters.

6) A sporting goods manufacturer budgets production of 45,000 pairs of ski boots in the first quarter and 30,000 pairs in the second quarter of the upcoming year. Each pair of boots require 2 kg of a key raw material. The company aims to end each quarter with ending raw materials inventory equal to 20% of the following quarter's material needs. Beginning inventory for this material is 18,000 kg and the cost per kg is $8. What is the budgeted materials purchases cost for the first quarter?

A) $720,000.
B) $672,000.
C) $576,000.
D) $729,600.
E) $864,000.

7) Alliance Company's budgets production of 24,000 units in January and 28,000 units in the February. Each finished unit requires 4 pounds of raw material K that costs $2.50 per pound. Each month's ending raw materials inventory should equal 40% of the following month's budgeted materials. The January 1 inventory for this material is 38,400 pounds. What is the budgeted materials need in pounds for January?
A) 102,400 pounds.
B) 96,000 pounds.
C) 57,600 pounds.
D) 140,800 pounds.
E) 83,200 pounds.

8) The sales budget for Modesto Corp. shows that 20,000 units of Product A and 22,000 units of Product B are going to be sold for prices of $10 and $12, respectively. The desired ending inventory of Product A is 20% higher than its beginning inventory of 2,000 units. The beginning inventory of Product B is 2,500 units. The desired ending inventory of B is 3,000 units. Total budgeted sales of both products for the year would be:
A) $42,000.
B) $200,000.
C) $264,000.
D) $464,000.
E) $500,000.

9) Western Company is preparing a cash budget for June. The company has $12,000 cash at the beginning of June and anticipates $30,000 in cash receipts and $34,500 in cash disbursements during June. Western Company has an agreement with its bank to maintain a minimum cash balance of $10,000. As of May 31, the company owes $15,000 to the bank. To maintain the $10,000 required balance, during June the company must:
A) Borrow $4,500.
B) Borrow $2,500.
C) Borrow $10,000.
D) Repay $7,500.
E) Repay $2,500.

10) Frankie's Chocolate Co. reports the following information from its sales budget:

Expected Sales:	July	$ 90,000
	August	104,000
	September	120,000

Cash sales are normally 25% of total sales and all credit sales are expected to be collected in the month following the date of sale. The total amount of cash expected to be received from customers in September is:

A) $30,000. B) $78,000.
C) $108,000. D) $120,000. E) $130,500.

11) Junior Snacks reports the following information from its sales budget:

Expected Sales:	October	$ 143,000
	November	151,000
	December	187,000

All sales are on credit and are expected to be collected 40% in the month of sale and 60% in the month following sale. The total amount of cash expected to be received from customers in November is:

A) $146,200. B) $85,800.
C) $151,000. D) $236,800. E) $60,400.

12) Justin Company's budget includes the following credit sales for the current year:
September, $25,000; October, $36,000; November, $30,000; December, $32,000. Experience has shown that payment for the credit sales is received as follows: 15% in the month of sale, 60% in the first month after sale, 20% in the second month after sale, and 5% is uncollectible. How much cash can Justin Company expect to collect in November as a result of current and past credit sales?

A) $19,700. B) $28,500.
C) $30,000. D) $31,100. E) $33,900.

13) Funcycle Manufacturing's budget includes the following credit sales for the current year: September, $145,000; October, $136,000; November, $120,000; December, $157,000. Experience has shown that payment for the credit sales is received as follows: 15% in the month of sale, 50% in the first month after sale, and 35% in the second month after sale. What are the cash collections of credit sales in the month of December?
A) $23,550.
B) $107,600.
C) $83,550.
D) $157,000.
E) $131,150.

14)Walter Enterprises expects its September sales to be 20% higher than its August sales of $150,000. Purchases were $100,000 in August and are expected to be $120,000 in September. All sales are on credit and are collected as follows: 30% in the month of the sale and 70% in the following month. Merchandise purchases are paid as follows: 25% in the month of purchase and 75% in the following month. The beginning cash balance on September 1 is $7,500. The ending cash balance on September 30 would be:
A) $31,500. B) $67,500. C) $54,000.
D) $61,500. E) $136,500.

15) The Gardner Company expects sales for October of $248,000. Experience suggests that 45% of sales are for cash and 55% are on credit. The company collects 50% of its credit sales in the month of sale and 50% in the month following sale. Budgeted Accounts Receivable on September 30 is $67,000. What is the amount of Accounted Receivables on the October 31 budgeted balance sheet?
A) $111,600.
B) $124,000.
C) $67,000.
D) $68,200.
E) $136,400.

16) Trago Company manufactures a single product and has a JIT policy that ending inventory must equal 5% of the next month's sales. It estimates that May's ending inventory will consist of 14,000 units. June and July sales are estimated to be 280,000 and 290,000 units, respectively. Trago assigns variable overhead at a rate of $1.80 per unit of production. Fixed overhead equals $400,000 per month. Compute the number of units to be produced and use this amount to compute the total budgeted overhead that would appear on the factory overhead budget for the month of June.

A) $920,200.
B) $904,900.
C) $930,100.
D) $922,000.
E) $878,800.

17) Zhang Industries budgets production of 300 units in June and 310 units in July. Each unit requires 1.5 hours of direct labor. The direct labor rate is $14 per hour. The indirect labor rate is $21.00 per hour. Compute the budgeted direct labor cost for July.

A) $6,300.
B) $6,510.
C) $9,450.
D) $9,765.
E) $16,275.

18) Zhang Industries is preparing a cash budget for June. The company has $25,000 cash at the beginning of June and anticipates $95,000 in cash receipts and $111,290 in cash disbursements during June. The company has no loans outstanding on June 1. Compute the amount the company must borrow, if any, to maintain a $20,000 cash balance.

A) $28,710.
B) $12,290.
C) $16,290.
D) $11,290.
E) $6,290.

19) Webster Corporation's budgeted sales for February are $325,000. Webster pays sales representatives a commission of 6% of sales dollars. The company pays a sales manager a monthly salary of $4,400 and expects advertising expense of $2,000 per month.1000 dep Compute the total selling expenses to be reported on the selling expense budget for the month of February.

A) $19,500. B) $6,400. C) $23,900.
D) $25,900. E) $21,500.

20) Webster Corporation is preparing a master budget for the first quarter of the year. The company budgets production of 2,680 units in January, 2,600 units in February and 2,740 units in March. Each unit requires 0.6 hours of direct labor. The direct labor rate is $12 per hour. Compute the budgeted direct labor cost for the first quarter budget.

A) $56,160. B) $57,744. C) $96,240.
D) $93,600. E) $48,120.

21)

Estimated selling price		$98	/unit
Budgeted unit sales, February		11,000	
Raw materials requirement per unit of output		5	pounds
Raw materials cost	$	3.00	per pound
Direct labor requirement per unit of output		2.5	direct labor-hours
Direct labor wage rate	$	18.00	per direct labor-hour
Predetermined overhead rate (all variable)	$	11.00	per direct labor-hour
Variable selling and administrative expense	$	2.70	per unit sold
Fixed selling and administrative expense	$	80,000	per month

The estimated net operating income (loss) for February is closest to:

A $5,800 B) $42,000
C) $35,500 D) $85,800

22) Wasilko Corporation produces and sells one product:
a. The budgeted selling price per unit is $114. Budgeted unit sales for February is 9,900 units.
b. Each unit of finished goods requires 6 pounds of raw materials. The raw materials cost $4.00 per pound.
c. The direct labor wage rate is $24.00 per hour. Each unit of finished goods requires 2.4 direct labor-hours.
d. Manufacturing overhead is entirely variable and is $9.00 per direct labor-hour.
e. The variable selling and administrative expense per unit sold is $1.60. The fixed selling and administrative expense per month is $70,000.

The estimated net operating income (loss) for February is closest to:

A) $50,000 B) $91,080
C) $21,080 D) $36,920

23-Todd Enterprises is preparing a cash budget for the second quarter of the coming year. The following data have been forecasted:

	April	May
Sales ……………………………………………….	$150,000	$157,500
Merchandise purchases ……………………………	107,000	112,400
Operating expenses:		
Payroll ………………………………………..	13,600	14,280
Advertising …………………………………..	5,400	5,700
Rent ……………………………………………..	2,500	2,500
Depreciation …………………………………	7,500	7,500
End of April balances:		
Cash ……………………………………………..	30,000	
Bank loan payable ………………………………	26,000	

Additional data:
(1) Sales are 40% cash and 60% credit. The collection pattern for credit sales is 50% in the month following the sale and 50% in the month thereafter. Total sales in March were $125,000.
(2) Purchases are all on credit, with 40% paid in the month of purchase and 60% paid in the following month.
(3) Operating expenses are paid in the month they are incurred.

(4) A minimum cash balance of $25,000 is required at the end of each month.
(5) Loans are used to maintain the minimum cash balance. At the end of each month, interest of 1% per month is paid on the outstanding loan balance as of the beginning of the month. Repayments are made at the end of the month if the cash balance exceeds $25,000.

Prepare the company's cash budget for May. Show the ending loan balance at May 31.

Multiple Choice Questions For Budget Cases :

1-Aloan Co. provides the following sales forecast for the next three months:

	January	February	March
Sales units	3,000	4,200	5,000

Units to be produced= 4200 + 500 (5 next month) - beg 420 (% 420)

The company wants to end each month with ending finished goods inventory equal to 10% of the next month's sales. Finished goods inventory on December 31 is 300 units. The budgeted production units for February are:

A) 5,000 units.
B) 4,200 units.
C) 4,700 units.
D) 4,120 units.
E) 4,280 units.

Answer: E

Explanation: February units + 10% of March units - January 31 inventory = February production

4,200 units + (5,000 units * .10) - (4,200 units * .10) = 4,280 units

2) Flack Corporation, a merchandiser, provides the following information for its December budgeting process:

The November 30 inventory was 1,800 units.
Budgeted sales for December are 4,000 units.
Desired December 31 inventory is 2,840 units.

Budgeted purchases are:

A) 5,040 units.
B) 1,240 units.
C) 6,840 units.
D) 4,000 units.
E) 5,800 units.

Answer: A

Explanation: Budgeted sales units + desired ending inventory - beginning inventory = purchases

4,000 units + 2,840 units - 1,800 units = 5,040 units

3- A department store has budgeted sales of 12,000 men's suits in September. Management wants to have 6,000 suits in inventory at the end of the month to prepare for the winter season. Beginning inventory for September is expected to be 4,000 suits. What is the dollar amount of the purchase of suits if each suit has a cost of $75.

A) $750,000.

B) $900,000.

C) $1,050,000.

D) $1,200,000.

E) $1,350,000.

Answer: C

Explanation: Budgeted purchases = Budgeted sales units + ending inventory - beginning inventory

Budgeted purchases = 12,000 + 6,000 - 4,000 = <u>14,000 suits</u>

14,000 suits * $75/suit = $1,050,000

4) A sporting equipment store expects to purchase $8,000 of ski boots in October. The store had $2,000 of ski boots in merchandise inventory at the beginning of October, and expects to have $3,000 of ski boots in merchandise inventory at the end of October to cover part of anticipated November sales. What is the budgeted cost of goods sold for October?

A) $5,000.

B) $7,000.

C) $8,000.

D) $9,000.

E) $10,000.

Units to be purchased= Units to be sold 7000+ Ending 3000 – Beginning2000

Cost of units purchased 8000= Cost of Goods sold + cost of ending3000 – cost of beginning 2000

COGS 7000 ?????= Beginning inventory 2000 +units produced 8000 - Ending inventory 3000

Answer: B

Explanation: Cost of Goods Sold = Beginning inventory + purchases - ending inventory

$2,000 + $8,000 - $3,000 = $7,000

5) Masterson Company's budgeted production calls for 56,000 liters in April and 52,000 liters in May of a key raw material that costs $1.85 per liter. Each month's ending raw materials inventory should equal 30% of the following month's budgeted materials. The April 1 inventory for this material is 16,800 liters. What is the budgeted materials need in liters for April?

A) 71,600 liters.

B) 39,200 liters.

C) 57,600 liters.

D) 56,000 liters.

E) 54,800 liters.

Answer: E

Explanation: Materials needed + ending inventory requirements - beginning inventory available = materials to be purchased

56,000 + (52,000 * 30%) - 16,800 = 54,800 liters

6) A sporting goods manufacturer budgets production of 45,000 pairs of ski boots in the first quarter and 30,000 pairs in the second quarter of the upcoming year. Each pair of boots require 2 kg of a key raw material. The company aims to end each quarter with ending raw materials inventory equal to 20% of the following quarter's material needs. Beginning inventory for this material is 18,000 kg and the cost per kg is $8. What is the budgeted materials purchases cost for the first quarter?

A) $720,000.

B) $672,000.

C) $576,000.

D) $729,600.

E) $864,000.

Answer: B

Explanation: Budgeted production units * materials requirement per unit =

materials needed
Materials needed + ending inventory requirements - beginning inventory available = materials to be purchased
Materials to be purchased * purchase price per unit - cost of materials purchases
45,000 * 2 kg = 90,000 kg; 90,000 kg + (30,000 * 2 kg * 20%) - 18,000 kg = 84,000 kg
84,000 kg * $8 = $672,000

7) Alliance Company's budgets production of 24,000 units in January and 28,000 units in the February. Each finished unit requires 4 pounds of raw material K that costs $2.50 per pound. Each month's ending raw materials inventory should equal 40% of the following month's budgeted materials. The January 1 inventory for this material is 38,400 pounds. What is the budgeted materials need in pounds for January?
A) 102,400 pounds.
B) 96,000 pounds.
C) 57,600 pounds.
D) 140,800 pounds.
E) 83,200 pounds.

Answer: A
Explanation: Budgeted production units * materials requirement per unit = materials needed
Materials needed + ending inventory requirements - beginning inventory available = materials to be purchased
24,000 * 4 lbs. = 96,000 lbs.
96,000 lbs. + (28,000 * 4 lbs. * 40%) - 38,400 lbs. = 102,400 lbs.

8) The sales budget for Modesto Corp. shows that 20,000 units of Product A and 22,000 units of Product B are going to be sold for prices of $10 and $12, respectively. The desired ending inventory of Product A is 20% higher than its beginning inventory of 2,000 units. The beginning inventory of Product B is 2,500 units. The desired ending inventory of B is 3,000 units. Total budgeted sales of both products for the year would be:
A) $42,000.
B) $200,000.

C) $264,000.
D) $464,000.
E) $500,000.

Answer: D
Explanation: (20,000 * $10) + (22,000 * $12) = $464,000

12000+ 30000 =42000-34500= 7500
9) Western Company is preparing a cash budget for June. The company has $12,000 cash at the beginning of June and anticipates $30,000 in cash receipts and $34,500 in cash disbursements during June. Western Company has an agreement with its bank to maintain a minimum cash balance of $10,000. As of May 31, the company owes $15,000 to the bank. To maintain the $10,000 required balance, during June the company must:
A) Borrow $4,500.
B) Borrow $2,500.
C) Borrow $10,000.
D) Repay $7,500.
E) Repay $2,500.

Answer: B
Explanation:

Beginning cash balance	$ 12,000
Add cash receipts	30,000
Less cash disbursements	(34,500)
Cash balance before financing	$ 7,500
Desired cash balance	10,000
Amount to borrow	$ 2,500

10) Frankie's Chocolate Co. reports the following information from its sales budget:

Expected Sales:	July	$ 90,000
	August	104,000

September	120,000

Cash sales are normally 25% of total sales and all credit sales are expected to be collected in the month following the date of sale. The total amount of cash expected to be received from customers in September is:
A) $30,000.
B) $78,000.
C) $108,000.
D) $120,000.
E) $130,500.

Answer: C
Explanation:

September cash sales (25% * $120,000)	$ 30,000
August credit sales (75% * $104,000)	78,000
Cash collected in September	$ 108,000

11) Junior Snacks reports the following information from its sales budget:

Expected Sales:	October	$ 143,000
	November	151,000
	December	187,000

All sales are on credit and are expected to be collected 40% in the month of sale and 60% in the month following sale. The total amount of cash expected to be received from customers in November is:
A) $146,200.
B) $85,800.
C) $151,000.
D) $236,800.
E) $60,400.

Answer: A
Explanation:

October credit sales collected (60% * $143,000)	$ 85,800
November credit sales collected (40% * $151,000)	60,400
Cash collected in November	$ 146,200

12) Justin Company's budget includes the following credit sales for the current year: September, $25,000; October, $36,000; November, $30,000; December, $32,000. Experience has shown that payment for the credit sales is received as follows: 15% in the month of sale, 60% in the first month after sale, 20% in the second month after sale, and 5% is uncollectible. How much cash can Justin Company expect to collect in November as a result of current and past credit sales?
A) $19,700.
B) $28,500.
C) $30,000.
D) $31,100.
E) $33,900.

Answer: D
Explanation:

15% of November sales (15% * $30,000)	$ 4,500
60% of October sales (60% * $36,000)	21,600
20% of September sales (20% * $25,000)	5,000
Total cash receipts	$ 31,100

13) Funcycle Manufacturing's budget includes the following credit sales for the current year: September, $145,000; October, $136,000; November, $120,000; December, $157,000. Experience has shown that payment for the credit sales is received as follows: 15% in the month of sale, 50% in the first month after sale, and 35% in the second month after sale. What are the cash collections of credit sales in the month of December?
A) $23,550.
B) $107,600.
C) $83,550.
D) $157,000.
E) $131,150.

Answer: E

Explanation:

15% of December sales (15% * $157,000)	$ 23,550
50% of November sales (50% * $120,000)	60,000
35% of October sales (35% * $136,000)	47,600
Total cash receipts	$ 131,150

Sept 150,000 +(150,000 x0.2) = 180000 x 30%=collections 54000 Sep + [150,000 x70%=105K]
September Payments = August 100,000 x 75% + September 120,000 x25% = 105000
Beg 7500 + 159000 – 105000= 61500

14)Walter Enterprises expects its September sales to be 20% higher than its August sales of $150,000. Purchases were $100,000 in August and are expected to be $120,000 in September. All sales are on credit and are collected as follows: 30% in the month of the sale and 70% in the following month. Merchandise purchases are paid as follows: 25% in the month of purchase and 75% in the following month. The beginning cash balance on September 1 is $7,500. The ending cash balance on September 30 would be:

A) $31,500. B) $67,500. C) $54,000.
D) $61,500. E) $136,500.

Answer: D
Explanation:

Cash balance, September 1		$ 7,500
Add cash receipts:		
30% of September sales (30% * $150,000 * 120%)	$ 54,000	
70% of August sales (70% * $150,000)	105,000	
Total cash receipts		159,000
Deduct cash disbursements:		
25% of September purchases (25% * 120,000)	$ 30,000	
75% of August purchases (75% * $100,000)	75,000	

Total cash disbursements	(105,000)
Cash balance, September 30	$ 61,500

15) The Gardner Company expects sales for October of $248,000. Experience suggests that 45% of sales are for cash and 55% are on credit. The company collects 50% of its credit sales in the month of sale and 50% in the month following sale. Budgeted Accounts Receivable on September 30 is $67,000. What is the amount of Accounted Receivables on the October 31 budgeted balance sheet?
A) $111,600.
B) $124,000.
C) $67,000.
D) $68,200.
E) $136,400.
Answer: D
Explanation: October credit sales not collected in October (55% * 248,000 * 50%) = 68,200

16) Trago Company manufactures a single product and has a JIT policy that ending inventory must equal 5% of the next month's sales. It estimates that May's ending inventory will consist of 14,000 units. June and July sales are estimated to be 280,000 and 290,000 units, respectively. Trago assigns variable overhead at a rate of $1.80 per unit of production. Fixed overhead equals $400,000 per month. Compute the number of units to be produced and use this amount to compute the total budgeted overhead that would appear on the factory overhead budget for the month of June.
A) $920,200.
B) $904,900.
C) $930,100.
D) $922,000.
E) $878,800.
Answer: B
Explanation: Budgeted ending inventory = 290,000 * 5% = 14,500 units
14,500 + 280,000 - 14,000 = 280,500 produced
280,500 * $1.80 = $504,900 + $400,000 = $904,900

17) Zhang Industries budgets production of 300 units in June and 310 units in July. Each unit requires 1.5 hours of direct labor. The direct labor rate is $14 per hour. The indirect labor rate is $21.00 per hour. Compute the budgeted direct labor cost for July.
A) $6,300.

B) $6,510.
C) $9,450.
D) $9,765.
E) $16,275.

Answer: B
Explanation: 310 * 1.5 * $14 = $6,510

18) Zhang Industries is preparing a cash budget for June. The company has $25,000 cash at the beginning of June and anticipates $95,000 in cash receipts and $111,290 in cash disbursements during June. The company has no loans outstanding on June 1. Compute the amount the company must borrow, if any, to maintain a $20,000 cash balance.
A) $28,710.
B) $12,290.
C) $16,290.
D) $11,290.
E) $6,290.

19) Webster Corporation's budgeted sales for February are $325,000. Webster pays sales representatives a commission of 6% of sales dollars. The company pays a sales manager a monthly salary of $4,400 and expects advertising expense of $2,000 per month. Compute the total selling expenses to be reported on the selling expense budget for the month of February.
A) $19,500. B) $6,400. C) $23,900.
D) $25,900. E) $21,500.

Answer: D
Explanation: Projected commission = $325,000 * 6% = $19,500
Total selling expenses = $19,500 + $4,400 + $2,000 = $25,900

20) Webster Corporation is preparing a master budget for the first quarter of the year. The company budgets production of 2,680 units in January, 2,600 units in February and 2,740 units in March. Each unit requires 0.6 hours of direct labor. The direct labor rate is $12 per hour. Compute the budgeted direct labor cost for the first quarter budget.
A) $56,160. B) $57,744. C) $96,240.
D) $93,600. E) $48,120.

Answer: B
Explanation:
January 2,680 * 0.6 * $12 = $19,296
February 2,600 * 0.6 * $12 = $18,720
March 2,740 * 0.6 * $12 = $19,728
Quarter total $19,296 + $18,720 + $19,728 = $57,744

20) Webster Corporation is preparing a master budget for the first quarter of the year. The company budgets production of 2,680 units in January, 2,600 units in February and 2,740 units in March. Each unit requires 0.6 hours of direct labor. The direct labor rate is $12 per hour. Compute the budgeted direct labor cost for the first quarter budget.

A) $56,160.
B) $57,744.
C) $96,240.
D) $93,600.
E) $48,120.

Answer: B
Explanation:
January 2,680 * 0.6 * $12 = $19,296
February 2,600 * 0.6 * $12 = $18,720
March 2,740 * 0.6 * $12 = $19,728
Quarter total $19,296 + $18,720 + $19,728 = $57,744

21)

Estimated selling price		98	units
Budgeted unit sales, February		11,000	
Raw materials requirement per unit of output		5	pounds
Raw materials cost	$	3.00	per pound
Direct labor requirement per unit of output		2.5	direct labor-hours
Direct labor wage rate	$	18.00	per direct labor-hour
Predetermined overhead rate (all variable)	$	11.00	per direct labor-hour
Variable selling and administrative expense	$	2.70	per unit sold
Fixed selling and administrative expense	$	80,000	per month

The estimated net operating income (loss) for February is closest to:
A) $5,800
B) $42,000
C) $35,500
D) $85,800

Answer: A
Explanation: The estimated unit product cost is computed as follows:

Direct materials	5pounds	$ 3.00per pound	$ 15.00
Direct labor	2.5hours	$ 18.00per hour	45.00
Manufacturing overhead	2.5hours	$ 11.00per hour	27.50
Unit product cost			$ 87.50

The estimated selling and administrative expense for February is computed as follows:

Budgeted unit sales	11,000
Variable selling and administrative expense per unit	$ 2.70
Total variable selling and administrative expense	$ 29,700
Fixed selling and administrative expenses	80,000
Total selling and administrative expenses	$ 109,700

The estimated net operating income for February is computed as follows:

Total sales 11,000 units × $98 per unit	$1,078,000
Cost of goods sold 11,000 units × $87.50 per unit	962,500
Gross margin	115,500
Selling and administrative expenses	109,700
Net operating income	$ 5,800

Question 22

Wasilko Corporation produces and sells one product:

a. The budgeted selling price per unit is $114. Budgeted unit sales for February is 9,900 units.
b. Each unit of finished goods requires 6 pounds of raw materials. The raw materials cost $4.00 per pound.
c. The direct labor wage rate is $24.00 per hour. Each unit of finished goods requires 2.4 direct labor-hours.
d. Manufacturing overhead is entirely variable and is $9.00 per direct labor-hour.
e. The variable selling and administrative expense per unit sold is $1.60. The fixed selling and administrative expense per month is $70,000.

The estimated net operating income (loss) for February is closest to:
A) $50,000
B) $91,080
C) $21,080
D) $36,920
Answer: C